STPM Physics · Semester 2 · Chapter 13

Capacitors

13.1 Capacitance

Definitions & laws

- **Capacitance** of a capacitor is the ratio of the magnitude of the charge on either plate to the potential difference between the plates ($C = \frac{Q}{V}$).
- **One farad** is the capacitance of a capacitor that stores a charge of one coulomb when the potential difference across it is one volt ($1\ F = 1\ C\ V^{-1}$).
- The capacitance depends only on the geometry of the capacitor (area, separation, shape) and the dielectric between the plates; it does not depend on the charge stored or the potential difference applied.
- When a capacitor is connected to a battery the potential difference V is fixed; when it is isolated (disconnected) the charge Q remains constant.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Capacitance	$C = \frac{Q}{V}$	C = capacitance (F); Q = charge on one plate (C); V = potential difference across the plates (V)
Capacitance of an isolated sphere	$C = 4\pi\varepsilon_0 R$	Isolated conducting sphere; R = radius (m); ε_0 = permittivity of free space ($F\ m^{-1}$)
Parallel plate capacitor (air)	$C = \frac{\varepsilon_0 A}{d}$	A = area of one plate (m^2); d = plate separation (m)
Parallel plate capacitor (dielectric)	$C = \varepsilon_0 \varepsilon_r \frac{A}{d}$	ε_r = relative permittivity of the dielectric (no unit)
Unit of capacitance	$F = C\ V^{-1} = A^2\ s^4\ kg^{-1}\ m^{-2}$	Typical values are μF (10^{-6} F), nF (10^{-9} F) and pF (10^{-12} F)

- On a graph of Q against V (straight line through the origin) the gradient is C.

How to answer (MPM style)

- State $C = \frac{Q}{V}$, substitute with units converted (μF, μC, mV) and give the answer in F with the suitable prefix.
- For a change question, state first which quantity is constant: V constant (still connected to battery) or Q constant (isolated), then use $Q = CV$ or $V = \frac{Q}{C}$.

- For a definition write "ratio of charge on one plate to the potential difference across the plates", and do not omit the word ratio.
- For a graph question, label the axes and state that the gradient of the Q–V graph equals the capacitance.

Section A traps

Trap: Wrong dependence: options stating that C is proportional to Q or to V; C is constant for a given capacitor.

Trap: Constant quantity: with the battery disconnected, doubling C halves V; with the battery connected, doubling C doubles Q.

Trap: Sphere sharing: two identical spheres sharing charge keep the same capacitance; $C = 4\pi\varepsilon_0 R$ depends on R only.

Trap: Unit slip: a distractor writes the farad with A instead of A^2, or with s^2 instead of s^4; check using $F = C\,V^{-1}$.

Tricks

Trick: Q = CV is a direct proportion: if V rises by $\frac{20}{12}$, Q rises by the same factor, and C need not be found.

Trick: Inverse relation: for the same charge a larger V means a smaller capacitance, since $C = \frac{Q}{V}$.

13.2 Parallel Plate Capacitors

Definitions & laws

- For a parallel plate capacitor the capacitance is directly proportional to the area of the plates and inversely proportional to their separation.
- The electric field between the plates is uniform, $E = \frac{V}{d} = \frac{\sigma}{\varepsilon_0}$, directed from the positive plate to the negative plate; the edge effects are neglected when d is much smaller than the plate dimensions.
- If the battery stays connected V is constant, so Q = CV changes when d or A changes; if the capacitor is isolated Q is constant.
- A slab of dielectric that partly fills the gap in layers (parallel to the plates) forms capacitors in series; a slab covering part of the area forms capacitors in parallel.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Parallel plate capacitance (air)	$C = \frac{\varepsilon_0 A}{d}$	C = capacitance (F); A = area of one plate (m^2); d = plate separation (m); $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}
Capacitance with dielectric	$C = \varepsilon_0\varepsilon_r \frac{A}{d}$	Dielectric completely filling the gap; ε_r = relative permittivity (no unit)
Field between the plates	$E = \frac{V}{d} = \frac{Q}{\varepsilon_0 A}$	E = electric field strength (V m^{-1}); V = potential difference (V); Q = charge (C)
Layers in series	$\frac{1}{C} = \frac{d_1}{\varepsilon_0 A} + \frac{d_2}{\varepsilon_0\varepsilon_r A}$	d_1, d_2 = layer thicknesses (m); ε_r = relative permittivity of the second layer
Stored energy	$U = \frac{1}{2}CV^2$	U = energy stored (J); C = capacitance (F); V = potential difference (V)

How to answer (MPM style)

- Write $C = \frac{\varepsilon_0 A}{d}$, convert cm^2 to m^2 ($\times 10^{-4}$) and mm to m ($\times 10^{-3}$) before substituting, then answer in F or pF with 3 s.f.
- For change questions use $C \propto \frac{A}{d}$, state what is constant (V or Q) and then use $Q = CV$ or $V = \frac{Q}{C}$.
- For a layered capacitor find each layer's capacitance ($C = \varepsilon_0\varepsilon_r \frac{A}{d_{layer}}$) and combine in series.
- For field strength use $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$, or $E = \frac{V}{d}$, and state the units V m^{-1}.

Section A traps

Trap: Inverted dependence: C increases when d decreases or A increases; options with C increased by increasing d or removing the dielectric are wrong.

Trap: Battery connected: when d increases, C decreases and Q = CV decreases; when d doubles Q is halved, not doubled.

Trap: Area unit: 6.0 cm^2 = 6.0×10^{-4} m^2 (not 6.0×10^{-2} m^2); a missing conversion gives an answer wrong by a factor of 10^2 or 10^3.

Trap: Combined change: halving d and doubling A multiplies C by 4, not 2.

Tricks

Trick: Ratio method: $\frac{C'}{C} = \frac{A'}{A}\frac{d}{d'}$; the constants ε_0 cancel.

> **Trick:** Layered gap: each layer of thickness $\frac{d}{2}$ has twice the air capacitance, so in series the value is quickly found as $\frac{2C_0 \times 6C_0}{2C_0 + 6C_0} = 1.5C_0$ for $\varepsilon_r = 3$.

13.3 Dielectrics

Definitions & laws

- A **dielectric** is an insulating material which becomes polarised in an electric field.
- **Relative permittivity (dielectric constant)** ε_r is the ratio of the capacitance of a capacitor with the dielectric between the plates to its capacitance when there is a vacuum between them ($\varepsilon_r = \frac{C}{C_0} = \frac{\varepsilon}{\varepsilon_0}$).
- **Polarisation:** in the external field the molecules of the dielectric become aligned (polar molecules) or stretched (non-polar molecules), so that opposite induced charges appear on the faces of the dielectric.
- The induced charges produce an electric field opposite in direction to the applied field, so the net field in the dielectric is reduced ($E = \frac{E_0}{\varepsilon_r}$), the potential difference between the plates falls for the same charge and so the capacitance increases.
- **Dielectric breakdown** occurs when the field exceeds the dielectric strength of the material and the insulator begins to conduct.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Capacitance with dielectric	$C = \varepsilon_r C_0 = \varepsilon_0 \varepsilon_r \frac{A}{d}$	C = capacitance (F); C_0 = capacitance without dielectric (F); ε_r = relative permittivity (no unit); A = plate area (m^2); d = plate separation (m)
Isolated capacitor (Q constant): field and p.d.	$E = \frac{E_0}{\varepsilon_r}$; $V = \frac{V_0}{\varepsilon_r}$	Capacitor isolated at constant Q; E_0, V_0 = field strength (V m^{-1}) and p.d. (V) without dielectric
Connected to battery (V constant)	$Q = \varepsilon_r Q_0$; $U = \frac{1}{2}\varepsilon_r C_0 V^2 = \varepsilon_r U_0$; $E = \frac{V}{d}$ unchanged	Battery connected, V constant; Q_0 = charge without dielectric (C); U_0 = energy without dielectric (J); U = energy (J)
Isolated capacitor: stored energy	$U = \frac{Q^2}{2C} = \frac{U_0}{\varepsilon_r}$	Isolated, Q constant; U = energy (J); Q = charge (C); C = capacitance (F)
Maximum voltage	$V_{max} = E_{max}\, d$	V_{max} = maximum safe voltage (V); E_{max} = breakdown field strength (V m^{-1}); d = plate separation (m)

How to answer (MPM style)

- Explain a capacitance increase in three points: the dielectric is polarised, the induced charges produce a field opposing the original field, the net field and the potential difference fall for the same charge, so $C = \frac{Q}{V}$ increases.
- State at the start whether the capacitor is connected to the battery (V constant) or isolated (Q constant); the answer depends on it.
- For numerical work use $C' = \varepsilon_r C$, then $Q = C'V$ (connected) or $V' = \frac{Q}{C'}$ (isolated) and give the answer with unit.
- Use the word "partially" for the cancelling effect: the net field is reduced, not zero.

Section A traps

Trap: Isolated vs connected: inserting a dielectric into an isolated capacitor reduces V, E and U by ε_r; with the battery connected Q and U increase by ε_r and V and E stay the same.

Trap: Factor error: energy changes by ε_r, not ε_r^2; a ratio of 1 : 25 or 25 : 1 for $\varepsilon_r = 5.0$ is wrong.

Trap: Complete cancellation: options saying the net field in the dielectric is zero, or that the polarisation field reinforces the external field, are wrong.

Trap: Breakdown: V_{max} depends on E_{max} and d, not on ε_r; doubling E_{max} doubles V_{max}.

Tricks

Trick: Quick table: connected: V same, E same, Q×ε_r, U×ε_r. Isolated: Q same, V÷ε_r, E÷ε_r, U÷ε_r.

Trick: Use U = ½QV: if Q is constant, U follows V; if V is constant, U follows Q.

13.4 Capacitors in Series and in Parallel

Definitions & laws

- In a **parallel** combination all capacitors have the same potential difference across them, and the total charge is the sum of the charges on the capacitors.
- In a **series** combination every capacitor carries the same charge, and the total potential difference is the sum of the potential differences across the capacitors.
- The equivalent capacitance in parallel is the sum of the capacitances; in series the reciprocal of the equivalent capacitance is the sum of the reciprocals.
- In series the potential difference across a capacitor is inversely proportional to its capacitance, so the smaller capacitor has the larger potential difference.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Capacitors in parallel	$C = C_1 + C_2 + C_3 + \ldots$	C = total capacitance (F); C_1, C_2, C_3 = individual capacitances (F)
Capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots$	C = total capacitance (F); C_1, C_2, C_3 = individual capacitances (F)
Two capacitors in series	$C = \frac{C_1C_2}{C_1 + C_2}$	C_1, C_2 = capacitances (F)
Series charge and voltage sharing	$Q = CV$; $V_1 = \frac{VC_2}{C_1 + C_2}$; $V_2 = \frac{VC_1}{C_1 + C_2}$	Q = charge on each capacitor (C); V = total p.d. (V); V_1, V_2 = p.d. across C_1 and C_2 (V)
n identical capacitors	Parallel: nC; series: $\frac{C}{n}$	n = number of identical capacitors, each of capacitance C (F)
Total stored energy	$U_{total} = \frac{1}{2}C_{total}V^2$	U_{total} = energy of the whole arrangement (J); C_{total} = total capacitance (F); V = p.d. across the arrangement (V)

How to answer (MPM style)

- Reduce the network step by step: combine each series or parallel group, redraw, and finish with a single equivalent capacitance.
- For voltage division use $Q = C_{total} \times V$ (same for each series capacitor) and then $V_1 = \frac{Q}{C_1}$.
- Write each stage with units (µF) and give the final value to 3 s.f.
- State the rule used at each step: "same charge in series" or "same potential difference in parallel".

Section A traps

Trap: Inverted rules: the formulae for capacitors are opposite to those for resistors; a distractor adds the capacitances in series.

Trap: Series bound: the series equivalent must be smaller than the smallest capacitor; 4 µF and 6 µF in series give 2.4 µF, whereas 10 µF (the sum) or 24 µF (the product) is wrong.

Trap: Voltage division: in series the smaller capacitor has the larger share: 3 µF and 6 µF across 12 V give 8 V and 4 V, not 4 V and 8 V.

Trap: Charge in a mixed network: the charge on the single series capacitor equals the sum of the charges on the two parallel capacitors, not equal to each one.

Tricks

Trick: Bounds: parallel gives more than the largest value; series gives less than the smallest; use this to eliminate options.

Trick: Identical capacitors C: the possible equivalents are 3C, $\frac{C}{3}$, $\frac{3C}{2}$ and $\frac{2C}{3}$ only.

13.5 Energy Stored in a Charged Capacitor

Definitions & laws

- The **energy stored in a capacitor** is the work done in charging it, equal to the area under the graph of potential difference against charge.
- **Energy density** of the electric field is the energy stored per unit volume of the field between the plates.
- When a capacitor is charged from a battery through a resistor, the battery supplies energy QV, of which ½QV is stored in the capacitor and ½QV is dissipated as heat in the resistor, independent of R.
- When a charged capacitor is connected to an uncharged capacitor the charge is conserved but the total energy decreases; the energy lost is dissipated as heat (and radiation).

Formulae

Quantity / law	Formula	Symbols, units & conditions
Energy stored in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	U = energy (J); Q = charge (C); V = potential difference (V); C = capacitance (F)
Energy density	$u = U/(Ad) = \frac{1}{2}\varepsilon_0 E^2$	u = energy density (J m^{-3}); A = plate area (m^2); d = plate separation (m); E = field strength (V m^{-1}); ε_0 = permittivity of free space (F m^{-1})
Energy density in a dielectric	$u = \frac{1}{2}\varepsilon_0\varepsilon_r E^2$	ε_r = relative permittivity (no unit)
Work done by the battery	$W = QV = CV^2$	W = work done by the battery (J); Q = charge (C); V = p.d. (V); C = capacitance (F)
Parallel sharing	$V = \frac{Q_{total}}{C_1 + C_2}$	V = common p.d. after sharing (V); Q_{total} = total charge (C); C_1, C_2 = capacitances (F)

- $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}.

How to answer (MPM style)

- For the energy stored write $U = \frac{1}{2}CV^2$ (choose the form matching the data), substitute with SI units, and give the answer in J with 3 s.f.
- For the V–Q graph, sketch a straight line through the origin and shade the triangle, showing that the area ½QV equals the energy.

- For energy lost on connecting capacitors find the charge conserved, the final V, the final energy and subtract it from the initial energy.
- Explain a loss as "energy is dissipated as heat in the connecting wires/resistor".

Section A traps

Trap: Missing ½: U = QV is the energy supplied by the battery, not the energy stored.

Trap: Wrong power: $U \propto V^2$ (so $3V_0$ gives $9E_0$, not $3E_0$) and $U \propto Q^2$ at fixed C.

Trap: Energy conservation: two equal capacitors, one charged, then connected: the final energy is half the initial, not equal to it.

Trap: Sum of energies: the total energy of two separate capacitors is the sum of each $\frac{1}{2}CV^2$, not $\frac{1}{2}(C_1 + C_2)(V_1 + V_2)^2$.

Tricks

Trick: Choose the form of U: $\frac{1}{2}CV^2$ at constant V, $\frac{Q^2}{2C}$ at constant Q, and $\frac{1}{2}QV$ if both are known.

Trick: Fixed capacitor: ratio method, $\frac{U'}{U} = (\frac{V'}{V})^2$ or $(\frac{Q'}{Q})^2$.

13.6 Charging and Discharging of a Capacitor

Definitions & laws

- The **time constant** $\tau = RC$ is the time taken for the charge on a discharging capacitor to fall to $\frac{1}{e}$ (about 37%) of its initial value; for charging it is the time for the charge to reach 63% $(1 - \frac{1}{e})$ of its final value.
- During discharge the charge, potential difference and current decay exponentially with time; during charging the current decays exponentially while the charge and potential difference rise exponentially towards their final values.
- At the start of charging the capacitor has no potential difference, so the initial current is $I_0 = \frac{V_0}{R}$; when the capacitor is fully charged the current is zero and the potential difference across it equals the EMF.
- The time constant is increased by increasing R or C; the final charge depends on C and V_0 only.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Discharging a capacitor	$Q = Q_0e^{-t/RC}$; $V = V_0e^{-t/RC}$; $I = I_0e^{-t/RC}$	Q, V, I = charge (C), p.d. (V), current (A) at time t (s); Q_0, V_0, I_0 = initial values; $I_0 = \frac{V_0}{R}$; R = resistance (Ω); C = capacitance (F)
Charging a capacitor	$Q = Q_0(1 - e^{-t/RC})$; $V = V_0(1 - e^{-t/RC})$; $I = I_0e^{-t/RC}$	$Q_0 = CV_0$ = final charge (C); V_0 = final p.d. (V); I_0 = initial current (A); R = resistance (Ω); C = capacitance (F); t = time (s)
Time constant	$\tau = RC$	τ = time constant (s); R = resistance (Ω); C = capacitance (F); $e^{-1} = 0.368$, $e^{-2} = 0.135$
Time for charge to halve	$t_{½} = RC \ln 2 = 0.693RC$	$t_{½}$ = time for the charge to halve (s); R = resistance (Ω); C = capacitance (F)
Logarithmic graph (discharge)	Gradient of ln Q against t = $-\frac{1}{RC}$	ln Q against t is a straight line of gradient $-\frac{1}{RC}$ (discharge)

How to answer (MPM style)

- Sketch the graph with labelled axes: Q (or V, I) against t, exponential decay from the initial value that tends to zero, or growth from zero to the final value $Q_0 = CV_0$; mark τ.
- Write the equation, substitute $\frac{t}{RC}$ with RC in seconds (kΩ × μF = ms), evaluate the exponential and give the answer with 3 s.f.
- For a time, rearrange with natural logarithms, e.g. $t = -RC \ln\frac{V}{V_0}$ for discharge or $t = -RC \ln(1 - \frac{V}{V_0})$ for charging.
- State that the exponential never reaches zero in a finite time, and that about 5τ is taken as practically complete.

Section A traps

Trap: Meaning of τ: after $t = \tau$ the charge in discharge is $\frac{1}{e}$ (37%) of the initial charge, not zero or half; in charging the capacitor reaches 63%, not 37%.

Trap: Dependence on R: doubling R doubles the time; the initial current $\frac{V_0}{R}$ decreases but the final charge CV_0 is unchanged.

Trap: Two time constants: $\frac{Q}{Q_0} = e^{-2} = 0.135$ (about 14%); a distractor gives 0.37 or 0.74.

Trap: Half-life of the current: I falls to $\frac{I_0}{2}$ at $t = RC \ln 2$ (0.693RC), not $\frac{RC}{2}$ or RC.

Tricks

Trick: Percentages to recall: charging at τ = 63%, at 2τ = 86%; discharging at τ = 37%, at 2τ = 14%.

Trick: Unit check: kΩ × μF gives ms, kΩ × mF gives s, MΩ × μF gives s.